

Final Project

What mainly affects my mood?

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DGM 6108 Programming Foundations for Digital Media

DGM 6109 Lab for Programming Foundations

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Overview

The world goes round and there are many ways to live. I seek happiness every day, I go about my day, but many times my mood is never as good as expected. A depressed mood lowers life expectations, so I don't want to be unhappy for a moment. When the mood is low, eating good food, buying things, playing games, watching shows, going out to However, many times, these do not make me happy. I want to explore which are the factors that really make me happy and which ones will turn me on. If I find the "code" to happiness, I can take control of my mood and make each day positive and meaningful.

At the beginning of the project, I envisioned that the amount of food intake and the amount of time I put into makeup and dressing each day would affect my happiness. But soon, these data did not turn out as well as I had hoped, and they did not have a significant positive or negative effect on my happiness. After that, I thought that the more money I spent the happier I would be, but that didn't seem to hold entirely true. It may be that since cash is rarely used nowadays but rather swiped, swiping doesn't bring the feeling of money fading away.

During this period, I noticed a positive correlation between the number of tasks completed and happiness, and a significant negative correlation between recreational time at home and happiness. In addition, the variable of number of steps was also discovered, and surprisingly, this data had a positive impact on my happiness.

I had been struggling with how to visualize the percentages of the three main activities, and this data had been left unused. It wasn't until I came across a graph in an article that I realized this data item fit perfectly with the ternary graph. Although I struggled with how to go about drawing this graph (how to convert the coordinates) for some time after that, it finally worked. Anything you learn will be used sometime in the future! Thanks to the mathematical knowledge I once had, I was able to calculate the correct coordinate scale relationship and make the visualization without regrets.

Data Recorded

I collect a lot of data for my projects because I want to track the real factors that influence happy moods. I hand-write notes on my iPad's "Goodnotes" when events occur (e.g., after wearing makeup, after a meal, after completing a task), and then summarize them in an excel sheet before bedtime. Some data is automatically recorded by electronic devices (e.g. daily game time, daily drama time, daily steps, money spent), which I summarize in excel once a week.

How much food I ate: I recorded the number of plates I ate by recording how many plates I used for each meal, including the bowls used for soup, rice, and fruit. Since I basically do not eat any snacks as well as do not drink any drinks, so it is reasonable to record only three meals a day(in dishes).

How long it took to dress and makeup: I recorded the time between getting ready to go out and actually going out as the full time spent on makeup and dressing.(in minutes)

Money spent: I keep track of my daily financial spending through an online banking app(in dollars). Also, online banking will have specific items of spending, which I categorize as: food, groceries, and quality of life.

Time spent on entertainment: I use my iPad to watch shows and play games. The iPad automatically records the time spent on each software each day. I add them together to calculate the total entertainment time(in minutes).

Tasks done: I record the daily tasks on the iPad and draw a check mark for those completed. At the end of the day, I summarize the data in an excel sheet(in pieces).

Steps: Mobile Health automatically records the number of steps I take each day.(in steps)

Main activities: At the end of the day, I manually record the daily activity percentage on the iPad(0%-100%). The percentage of the three activities combined is 1.

Happy mood: I record my daily happy score on a 1-4 point scale. Where 2.5 is the middle value, increasing or decreasing as appropriate.

Since it was found through the preliminary scatter plots that the amount of food eaten and the time spent in make-up did not observe a clear relationship with my happy mood, I stopped collecting these data over a period of time.

Since, in later visualizations, it was also not obvious to observe the relationship between the money spent and my happy mood, I then stopped uploading this part of the data as well, although it will be continuously recorded in my online banking.

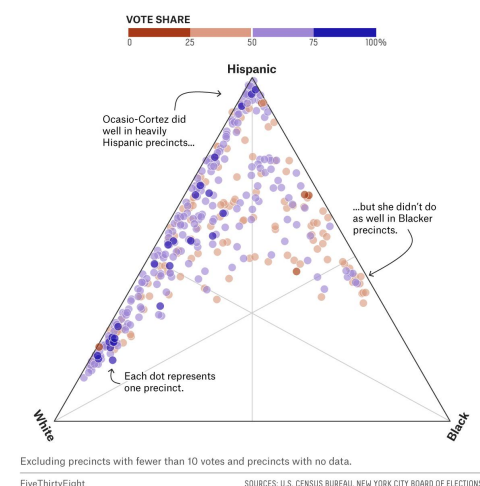
ALL DATA													
Date	Food I eat (dishes)	Time(min) of makeup	Money(\$) paid all	Money(\$)			Time(min) of entertain	Time(min) of videos	Time(min) of games	Done Tasks	Steps (step)	Main Activity (Study, Play, Goout)	Happy Mood (points)
				Meal	Grocery	Life Quality							
10 Oct 2022	9	15	101	19	29	53+51	100	100	0	3	8795	5,30,65	3.4
11 Oct 2022	5	20	18	11+5+28	2		70	30	40	4	5159	50,5,45	3.2
12 Oct 2022	5	5	13	13			200	0	200	2	187	40,45,5	2.6
13 Oct 2022	4	10	67	38+29			120	40	80	3	4723	50,20,30	3.4
14 Oct 2022	4	30	0		20+26	44 (-90)	80	20	60	3	7368	35,15,50	3.6
15 Oct 2022	4	20	199	47	16+22+12	32+50+20	120	80	40	2	3906	40,30,30	3.3
16 Oct 2022	6	10	22	22			90	60	30	1	45	40,50,10	1.6
17 Oct 2022	4	10	1			1	90	50	40	2	135	65,30,5	2.7
18 Oct 2022	6	25	971		8+35	15+913	190	70	120	3	4469	30,40,30	2.5
19 Oct 2022	7	5	29	29			240	80	160	1	47	25,70,5	2.1
20 Oct 2022	4	15	171	55	20+96		140	60	80	2	5693	25,35,40	3.3
21 Oct 2022	12	50	169	91	12	17+44+5	40	0	40	4	9689	20,15,65	3.8
22 Oct 2022	4	15	96	33		53+10	130	50	80	2	6165	25,30,45	3.6
23 Oct 2022	5	0	0				240	80	160	0	93	25,65,10	2
24 Oct 2022	7	10	131		22+109		300	100	200	0	1905	30,50,20	2.3
25 Oct 2022	6	30	102	16+6+30		(+50)	220	60	160	1	4585	20,50,30	1.7
26 Oct 2022	9	5	0				260	60	200	0	99	45,45,10	2.4
27 Oct 2022	13	15	106	67+23	23+33	(-40)	120	40	80	3	6656	40,25,35	3.5
28 Oct 2022	8	25	25	25		25	210	90	120	1	7921	2,53,45	1.4
29 Oct 2022	9	30	33		20(-7)+20		80	0	80	4	9183	5,20,75	3.8
30 Oct 2022	5	15	57		35+22		130	50	80	4	7620	30,20,50	3.1
31 Oct 2022	4	0	61	39		22	120	0	120	4	18	80,15,5	3.6
1 Nov 2022	6	20	46	30	16		120	40	80	3	3247	50,35,15	2.6
2 Nov 2022	8	0	0				200	80	120	1	16	50,45,5	1.4
3 Nov 2022	6	5	61	37	14+10		200	80	120	4	2719	35,45,20	1.8
4 Nov 2022	8	5	50	53	11+1	(-15)	160	0	160	4	4752	20,40,35	2.3
5 Nov 2022	6	15	91	13+6+72			120	0	120	2	2769	45,30,20	2.7
6 Nov 2022	5	10	112	38	60	14	80	0	80	4	6066	35,20,45	3.6
7 Nov 2022	4	5	71	11(+12)	10+7+31		160	80	80	3	1209	40,35,25	2.7
8 Nov 2022	4	10	0				100	60	40	4	4350	50,30,20	3.1
9 Nov 2022	7	0	48		48		110	70	40	2	6	60,38,2	2.6
10 Nov 2022	5	10	92	39		63	50	50	0	3	4009	55,10,35	3.1
11 Nov 2022	8	5	36		36		130	90	40	4	1110	30,35,35	3.3
12 Nov 2022			53		37+6+10		200	80	120	2	6697	40,40,20	2.3
13 Nov 2022			40	40			260	80	180	1	29	5,60,35	2.6
14 Nov 2022			12	12			70	30	40	4	129	50,25,25	3.2
15 Nov 2022			22	17+5			40	0	40	4	4335	65,25,10	3.6
16 Nov 2022			0				120	40	80	3	228	55,35,15	2.8
17 Nov 2022			50	36	14		160	40	120	4	5291	35,35,30	3.1
18 Nov 2022			78	15(+35)+10	7+11		120	50	210	2	3572	10,45,45	2.1
19 Nov 2022			50		50		140	40	100	2	7126	10,30,60	3.1
20 Nov 2022			16		3+13		320	60	260	1	1777	5,80,15	1.7
21 Nov 2022			42	13+29			160	40	120	2	863	40,40,20	2.2
22 Nov 2022			66	55		11	120	40	80	2	115	45,45,10	2.6
23 Nov 2022			70		59+11		80	40	40	3	6129	40,15,45	3.6
24 Nov 2022			37	37			120	40	80	2	3900	35,45,30	2.3
25 Nov 2022			10		10		150	30	120	2	2826	35,50,15	2.5
26 Nov 2022			67	57		10	160	40	120	4	5565	20,35,45	3.3
27 Nov 2022			60	60			150	30	120	2	306	40,50,10	3.1
28 Nov 2022			12	5	7		0	0	0	3	4082	68,2,30	3.7
29 Nov 2022							120	20	100	2	4893	60,20,20	3.4
30 Nov 2022							100	100	0	2	116	70,20,10	3.2
1 Dec 2022							150	150	0	2	9936	10,35,55	2.7
2 Dec 2022							60	0	60	4	5327	50,5,45	3.1
3 Dec 2022							0	0	0	4	4598	45,5,50	3.6
4 Dec 2022							70	20	50	2	1015	50,40,10	2.6
5 Dec 2022							140	140	0	3	236	60,35,5	2.8
6 Dec 2022							100	0	100	3	4977	60,20,20	3.4
7 Dec 2022							120	50	70	2	78	45,50,5	2.7
8 Dec 2022							90	0	90	3	6246	35,35,30	3.1
9 Dec 2022							200	0	200	3	4209	15,40,45	2.9
10 Dec 2022							190	0	190	3	5751	20,40,40	2.5
11 Dec 2022							230	30	200	1	96	20,70,10	1.9
12 Dec 2022							180	30	150	1	234	40,45,15	1.7
13 Dec 2022							100	0	100	3	5012	50,15,35	3.5
14 Dec 2022							110	30	80	1	71	65,33,2	1.5

Rational & Sketches

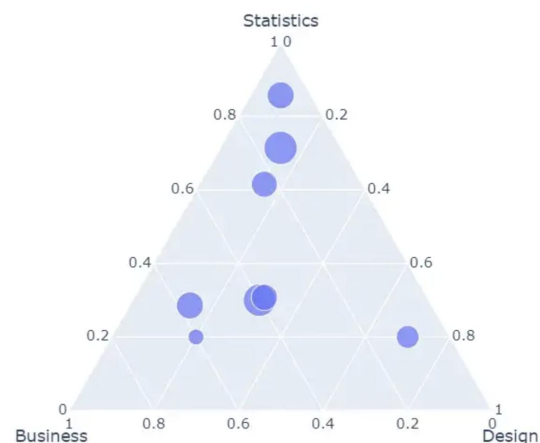
My final visualization is a combination of Sketch 1 and Sketch 2 designs.

First, the large triangular coordinate system shows very clearly the different percentages of the main activities of the day in each of my data items. This visualization fits my data very well and very effectively shows the impact of that data item on the overall results at different percentages. I was so intrigued by this representation when I read this web page that I referenced it in my final visualization. However, the axes of the different dimensions of the triangles in Idea 1 and Sketch 1 are defined by the heights of the triangles, and I felt that if I wrote the coordinate scales on the "heights", they would overwrite my data points and affect the reading. Therefore, in the final visualization, I chose the coordinate representation of Idea 2, using the sides of the triangles as the axes for each major activity, and drawing the axis scales and axis baselines to aid reading.

Idea 1:



Idea 2:



Link of Idea 1:

<https://fivethirtyeight.com/features/what-the-squad-tells-us-about-progressives-ability-to-win-voters-of-color/>

Link of Idea 2:

<https://towardsdatascience.com/basics-of-ternary-plots-with-pythons-plotly-e5990ea399a5>

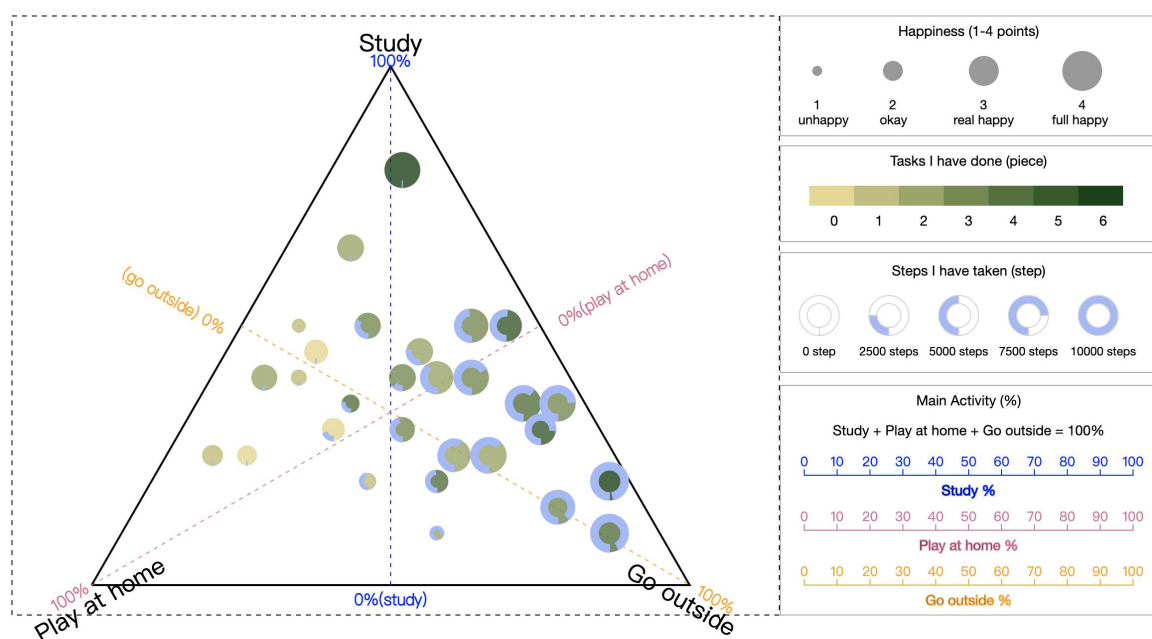
Secondly, the number of sectors in the circle was chosen to represent the number of tasks done during the day because it is not only clear, but also very aesthetically pleasing. Compared to some very subdivided data items (e.g., steps, money, happiness index, etc.), the number of tasks completed is integer and can be well represented in small images on a case-by-case basis. I have used different colors in sketch 1 to distinguish the number of completed tasks. However, if I use the same color gradient, the colors are very close to each other;

if I use different colors, I need to read back the description in the key repeatedly to determine the value size represented by that color. Neither of these two approaches was entirely satisfactory to me, and the number of sectors was a good way to visualize this data item.

Next, I chose to use a "paper airplane" to represent the number of steps in a day, with different colors and different rotation angles. Compared to the arcs in sketch 1, the "paper planes" with different colors and selected angles ensure that the data is very clear even with a small number of steps. Also, the "paper airplane" is lively, I don't want my visualization to be too rigid, I want it to be meaningful and interesting.

Finally, my happiness index is expressed in the final visualization in terms of both the size of the circle (radius) and the color. If only the radius is used, it is somewhat difficult to see the difference in size for a circle with an empty fill and only the border. If only color is used, similar data points could cause confusion. Therefore, gradient colors and gradient radius can be combined to help observe the different happiness scores.

Sketch 1:

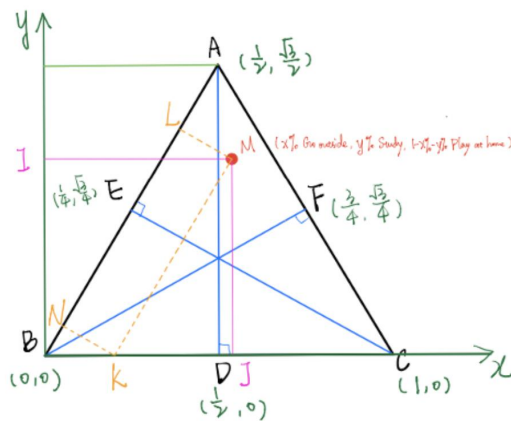


Link of Sketch 1:

<https://digitalmedia.neu.edu/6108maxiaoq/term/attempt1/final/>

Sketch 1 has the advantage that each data item is meaningful and the triangular coordinate system is distinctive. I can observe obvious patterns between the data through this sketch. The disadvantage is that when the number of steps is very small, the arc on the circle where that data item is located is almost invisible and very difficult to observe. Also, since there is no very detailed interval scale on the large triangle, it takes a lot of effort for a complete novice to understand how each dimension is represented.

In Sketch 1's big equilateral triangle, I cannot use the height of the triangle as the x/y axis to locate the data points. So I used the vertex of the lower left corner (Point B in the figure below) as the origin to draw the data points. This required me to figure out the coordinate values of each data point relative to this origin, so I used mathematical knowledge to figure out the true x-values of the data points.



$$\because \triangle MKJ \sim \triangle ABD$$

$$\therefore \frac{KJ}{BD} = \frac{MJ}{AD}$$

$$KJ = \frac{MJ \cdot BD}{AD} = \frac{\frac{y}{\sqrt{3}} \cdot \frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{y}{\sqrt{3}} \cdot \frac{1}{2} \cdot \frac{2}{\sqrt{3}} = \frac{y}{3}$$

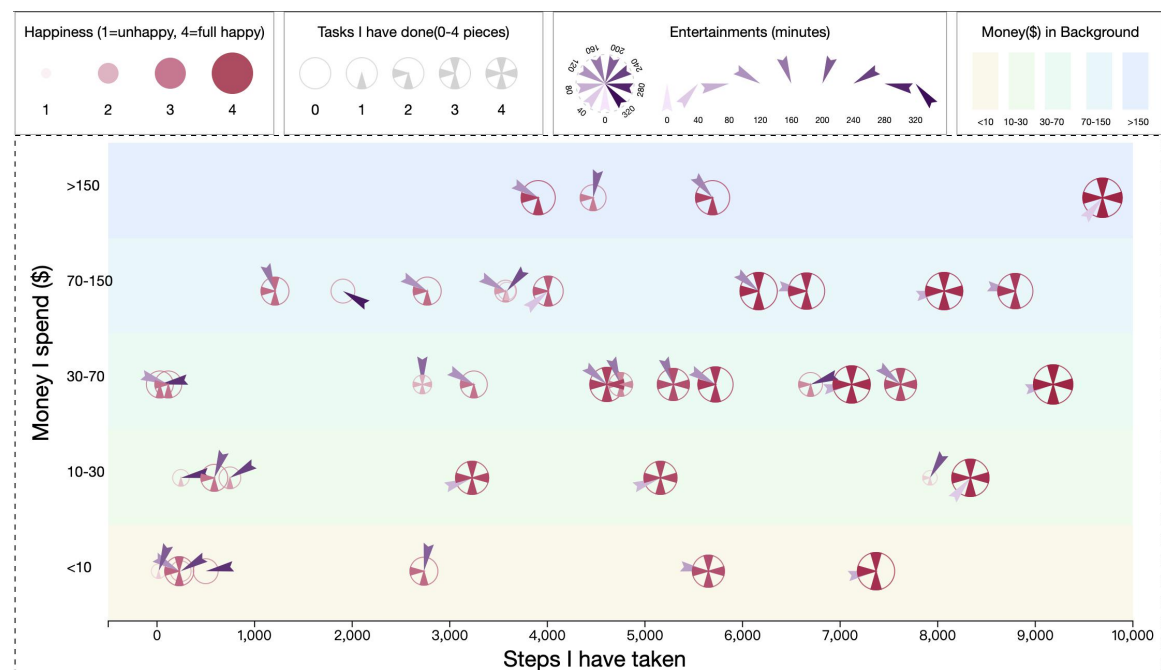
$$\because \triangle BNK \sim \triangle BEC$$

$$\therefore \frac{BK}{BC} = \frac{NK}{EC}$$

$$BK = \frac{NK \cdot BC}{EC} = \frac{x \cdot 1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}} x$$

$$\therefore M_x = BJ = BK + KJ = \frac{2}{\sqrt{3}} x + \frac{y}{\sqrt{3}} = \frac{2x+y}{\sqrt{3}}$$

Sketch 2:



Link of Sketch 2:

<https://digitalmedia.neu.edu/6108maxiaoq/term/attempt2/final/>

Sketch 2 has the advantage of rich colors and vivid images. In this sketch, I have implemented the number of sectors to show the number of tasks done and the rotating "paper planes" to show the number of steps. The disadvantage is that I found that the data item "Money" has little influence on my theme and there is no obvious pattern.

Analysis

I use visualization to explore what mainly affects my happy mood. My hypothesis is the more tasks I have done, the happier I am. Furthermore, the more steps I have taken, the happier I am. Additionally, playing at home has a negative impact on my happiness, while studying and going out both have positive impacts on my happiness.

Based on my data, I found that the factors that influence happiness are complex. Through visualization, I concluded that the number of steps, the number of tasks done, studying and going out had a positive impact on my daily happy mood, while playing at home (playing games and watching videos) had a negative impact on my happiness. In addition, the amount of time spent on makeup each day and the amount spent mostly acted on immediate mood and had an insignificant effect on overall mood; the amount of food had a positive effect, but the variation was small, so it was not appropriate for analysis. Although some factors show an overall positive effect on happiness, there are still exceptions that occur and the weight of negative effects from other factors needs to be taken into account.

The visualized conclusions shocked me because I had always thought that the time invested in makeup and the amount consumed would have a significant impact on my mood. Because I was happy every day when I put on makeup and had a strong sense of pleasure after spending money. But now I realize that these are the result of adrenaline action, which will give a lot of happy emotions for a short time, but after some time (an hour) has passed, these pleasures will disappear. True pleasure requires a lot of constant effort, such as continuous exercise, completing a task. These cannot be achieved overnight in a short period of time, but will bring a deep sense of satisfaction afterwards.

This project has changed my perception of the emotions of happiness, short-lived happiness and long-lasting happiness. Before, I always thought that when I relaxed, I played games, watched dramas, or watched movies. But now I feel that these do not bring me as much pleasure as I thought, or even have a negative impact on my mood, and that I should choose to go out and have a look around or learn something new. Learning and going out can bring mental or physical improvement, and these can only produce long-lasting happiness.

In this project I explored the effect of different factors on happiness, and the results were unexpected but supported my later hypothesis. I think that afterwards I can collect the change in mood before and after wearing makeup, and before and after consumption (in units of every 5 minutes), and this can analyze the change in happy mood more deeply. In the future, I would like to explore the duration of the effect of different factors on happiness. Everyone wants to be happier than the day before, but there is ambiguity about the mechanisms that generate happiness. Through this project, I have updated my

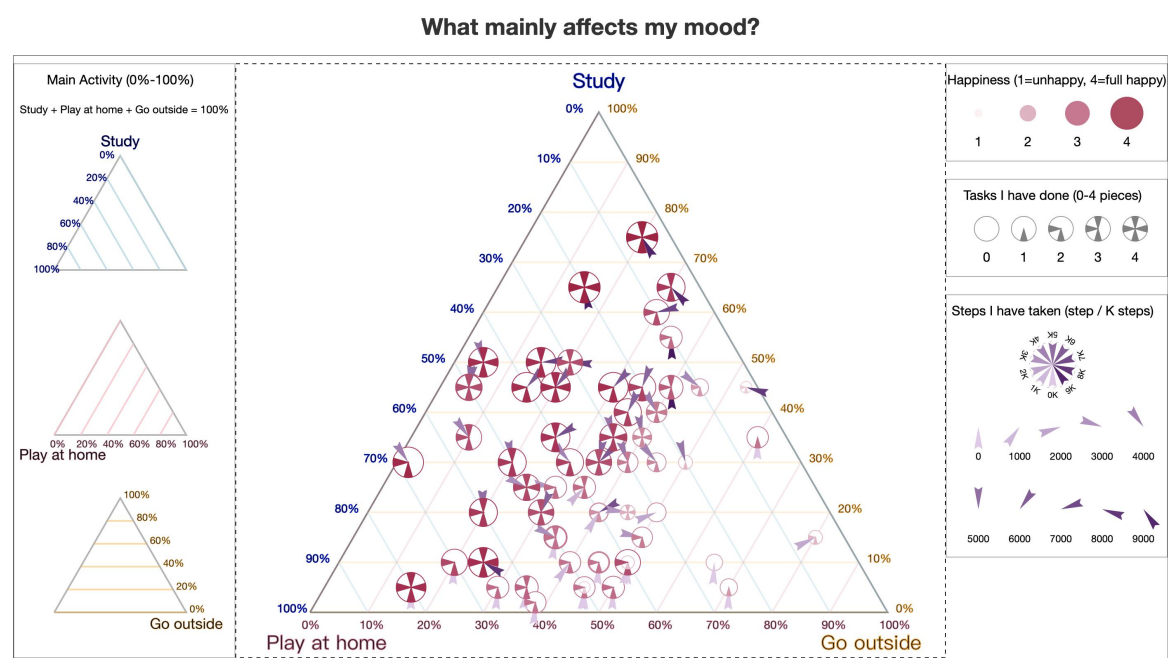
knowledge and hope to have a clearer grasp of the "code" for happiness in the future.

Before assigning this project, I had the habit of keeping a log, but would not organize and analyze this information recorded. I will plan tasks for each day and check off the ones that are completed, which is helpful for my project data. This project motivated me to collect more segmented data, such as happiness scores for every five minutes before and after wearing makeup. I want to use them to explore the duration of short-term pleasure (and long-term pleasure).

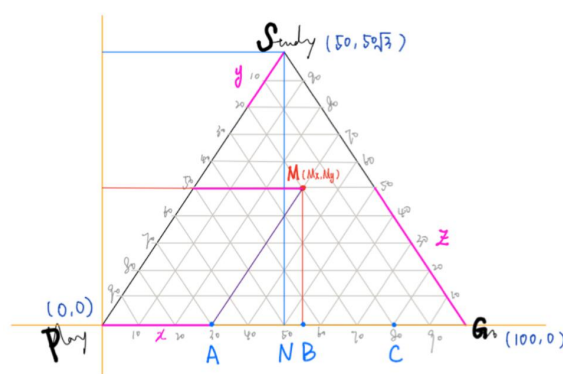
In addition, this project motivated me to collect the happiness index of people around me about the same factor. I think due to the individual differences of each person, different factors play a different role in happiness. For example, I am not very interested in food, so food brings me little emotional impact; but for my roommate, the abundance of food she eats every day will significantly affect her happiness. In this way, I would like to collect more data from different people to explore the effect of each factor on happiness in different types of people.

Final Visualization and Process

I use ternary scatter plots to present data visualization. Each side of the triangle is used to represent the scale of different activities. The percentage of the three main activities in a day added together is 1 (100%). In the large triangle, I used different colored lines to assist in distinguishing how each dimension (different main activities) was measured (blue for study, palevioletred for playing at home, orange for going outside). The number of steps is indicated synchronously by the size of the circle and the shade of red. The number of sectors in each data circle indicates the number of tasks done (0-4 pieces). The number of steps is shown simultaneously by the shade of purple and the angle of rotation of the "paper airplane".



Regarding how to calculate the x/y value coordinates of each data point relative to the canvas origin (point P in the figure below), I used some math to do the operation. This is different from the result of the calculation in my attempt 1, because I converted the base axis. In Attempt 1, the percentages for each main activity were taken using the heights of the triangles, while in the final project they were taken using the sides of the triangles.



$$\begin{aligned} \because \triangle MBC \sim \triangle SNG \\ \therefore \frac{MB}{SN} &= \frac{MC}{SG} \\ MB &= \frac{MC \cdot SN}{SG} \\ &= \frac{Z \cdot 50\sqrt{3}}{100} \\ &= \frac{\sqrt{3}}{2} Z \end{aligned}$$

$$\begin{aligned} \therefore M_y &= MB \\ &= \frac{\sqrt{3}}{2} Z \end{aligned}$$

$$\begin{aligned} \because \angle MAC &= \angle MCA \\ \therefore MA &= MC = Z \\ \because \triangle MAB \sim \triangle SPN \\ \therefore \frac{AB}{PN} &= \frac{MB}{SN} \\ AB &= \frac{MB \cdot PN}{SN} = \frac{MC}{SG} \cdot 50 \\ &= \frac{Z}{100} \cdot 50 \\ &= \frac{1}{2} Z \end{aligned}$$

$$\begin{aligned} \therefore M_x &= PA + AB \\ &= x + \frac{1}{2} Z \end{aligned}$$

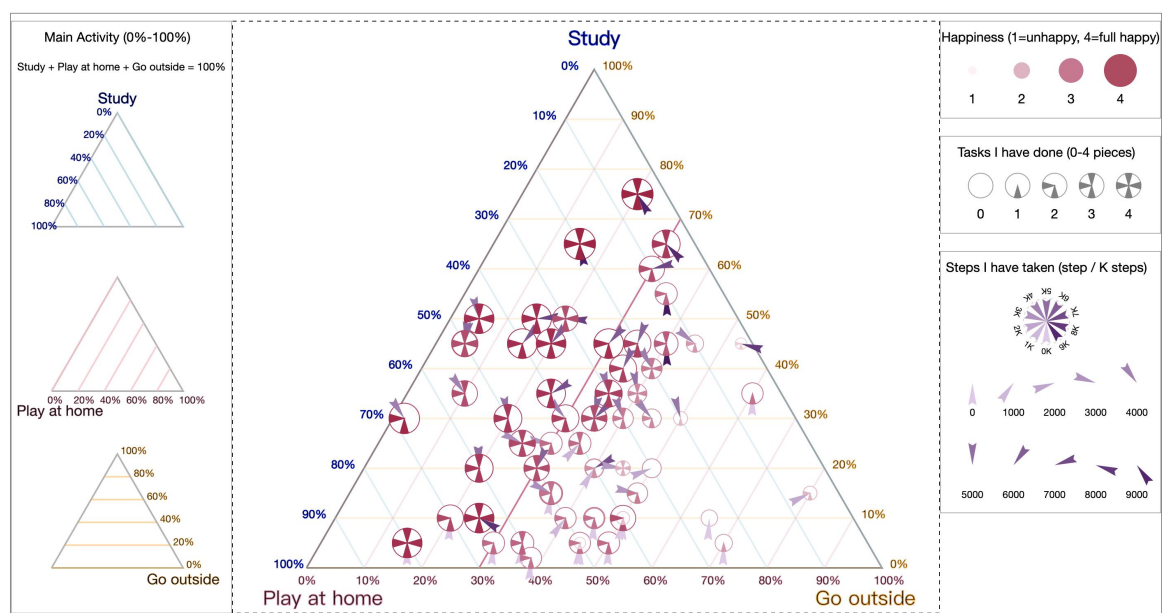
About how to indicate the number of tasks done with different number of sectors in programming. I envision drawing as many sectors (fans) as there are done tasks on that data point. But if I use this logic: first create a separate function for each number of done, and then draw each sector it needs in each corresponding function, I would have to write a lot of repetitive code. So I changed my drawing logic: first I treat each individual sector as a drawing unit, and then use the filtered data that needs to be drawn in the sector at that position. In other words, as long as the data points that need to be plotted in that sector, that sector is plotted.

```
function filterDoneOne(value) {
  return value.done == "1" || value.done == "2" || value.done == "3" || value.done == "4";
}
function filterDoneTwo(value) {
  return value.done == "2" || value.done == "3" || value.done == "4";
}
function filterDoneThree(value) {
  return value.done == "3" || value.done == "4";
}
function filterDoneFour(value) {
  return value.done == "4";
}

/**** Draw circle inner arc ****/
drawing.selectAll("path.fan1") //draw the first fan
  .data(data.filter(filterDoneOne))//When the value.done = 1 or 2 or 3 or 4 (excluded the case of 0),
  draw the first fan!
  .join("path")
  .attr("transform", function (value) { return borderXY(value) }) //get position of the center of the
  circle corresponding to the fan
  .attr("d", d3.arc()
    .innerRadius(0)// The inner radius of the fan is from 0(the center of the corresponding circle)
    .outerRadius(function (value) { return rScale(value.mood) }) // get the radius of the outer
    radius of the fan: use rScale function-->line:148
    .startAngle(2.86) // 2.86--3.14--3.42(0.28) = 0 degree
    .endAngle(3.42)
  )
}
```

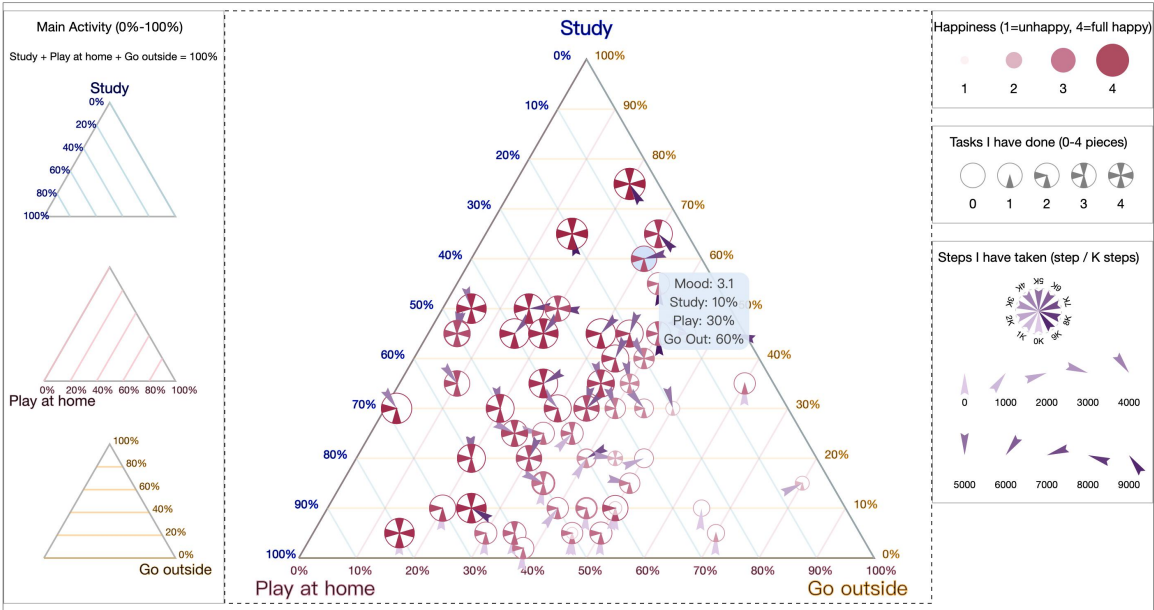
When the mouse is placed on a certain reference line, the color of that reference line deepens to assist in observing the difference between the data points on both sides.

What mainly affects my mood?



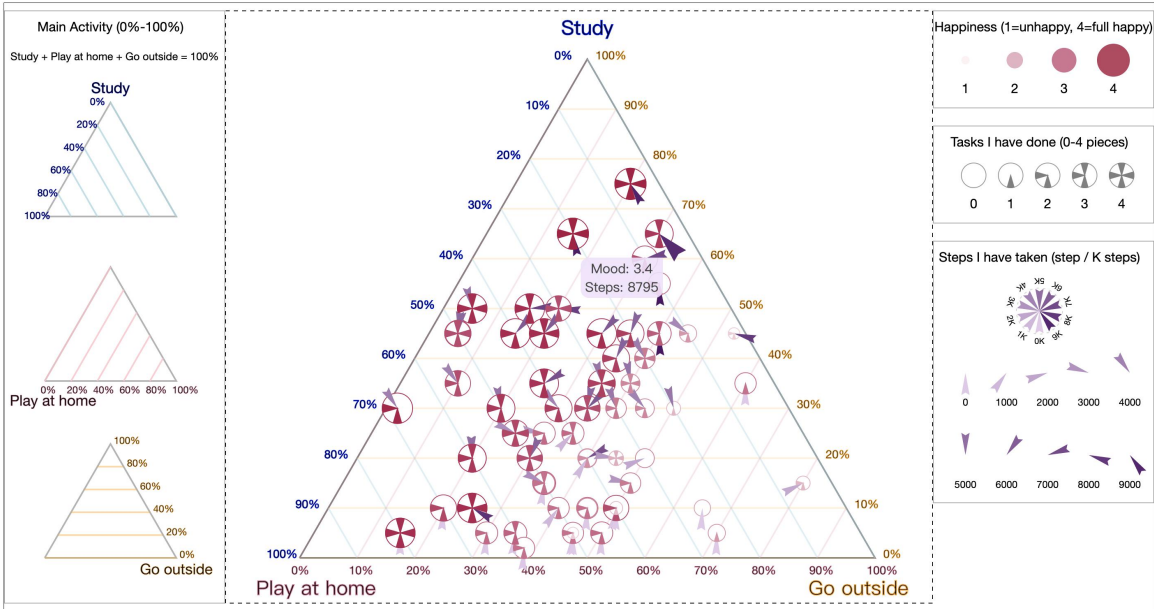
When the mouse is placed over the circle of a data point, the fill color of the circle turns blue and a text box appears on the right side showing the happy score and the separate percentages of the main activities represented by that data point.

What mainly affects my mood?



When the mouse is placed on the "paper airplane" in the circle of a data point, the paper airplane will become bigger and a text box will appear on the left side showing the happy score and the number of steps represented by that data point.

What mainly affects my mood?



Citations

Visualization ideas:

<https://fivethirtyeight.com/features/our-51-best-and-weirdest-charts-of-2021/>

<https://fivethirtyeight.com/features/what-the-squad-tells-us-about-progressives-ability-to-win-voters-of-color/>

<https://towardsdatascience.com/basics-of-ternary-plots-with-pythons-plotly-e5990ea399a5>

Tooltip of web interactivity:

<https://bl.ocks.org/d3noob/97e51c5be17291f79a27705cef827da2>

Color code:

<https://g.co/kgs/nt4PtA>